

PhishLens Project Documentation

Advanced Client-Side Phishing Detection & URL Intelligence Framework

1. Project Introduction

PhishLens ek modern, lightning-fast aur privacy-focused web application hai jo users ko deceptive URLs, malicious hyperlinks, aur phishing attacks se bachane ke liye design ki gayi hai. Aaj ke digital yug mein, phishing attacks social engineering ka sabse bada hathiyar ban chuke hain jo banks, credentials, aur personal data ko target karte hain. PhishLens ek transparent analysis engine provide karta hai jo kisi bhi link ko deep inspect karta hai bina user ke data ko kisi insecure external server par expose kiye.

Core Objective: Users ko link click karne se pehle uski asliyat dikhana, unhe deceptive homoglyphs se agyaat karna, aur ek real-time risk evaluation score provide karna.

2. Architecture & Technical Design

PhishLens ko ek lightweight, scalable, aur client-side architecture par banaya gaya hai:

- Pure Client-Side Engine:** Isme kisi heavy backend database ya complex server infrastructure ki zaroorat nahi padti. Saari parsing aur heuristics browser ke andar execute hoti hain.
- Tech Stack:** HTML5, modern CSS3 (Flexbox/Grid layouts), aur vanilla JavaScript, jisme AI-powered insights ke liye Gemini API integration shaamil hai.
- Deployment Ready:** Static files par based hone ke karan ise GitHub Pages, Netlify, ya Vercel par instantly deploy kiya ja sakta hai.

3. Target Audience & Use Cases

Yeh tool students, cybersecurity beginners, everyday web users, aur IT professionals ke liye extremely useful hai jo instant URL verification chahte hain.

4. Core Features In-Depth

Anatomy Engine

URL Anatomy Visualizer

Yeh feature kisi bhi lambe ya suspicious URL ko tod kar uske alag-alag components—jaise ki Protocol (HTTPS/HTTP), Subdomain, Registered Domain, Path, aur Query Parameters—mein clearly breakdown karta hai.

Security Guard

Punycode & Homoglyph Detection

Phishers aksar original domains ke jaise dikhne wale fake unicode/homoglyph domains (jaise `g00gle.com` ya internationalized characters) use karte hain. PhishLens in deceptive tricks ko turant detect karta hai.

Risk Scoring

Real-Time Risk Scorecard

Heuristic rules aur pattern analysis ke basis par, PhishLens ek dynamic risk score generate karta hai. Yeh score user ko batata hai ki link safe hai, suspicious hai, ya dangerous hai.

AI Dissection

Visual Threat & AI Dissection

Advanced threat patterns, urgency triggers, aur fake login indicators ko identify karne ke liye system deep heuristics aur Gemini API ka support leta hai taaki comprehensive insights mil saken.

5. Educational Impact & Conclusion

PhishLens sirf ek detection tool nahi hai, balki ek **educational platform** bhi hai. Jab user kisi URL ko analyze karta hai, toh woh samajhta hai ki attackers domain spoofing kaise karte hain. Isse digital literacy aur online safety strong hoti hai.

Summary: PhishLens aapke project ka ek behtareen representation hai jo web security aur user empowerment ka perfect balance pesh karta hai.